

Unraveling the Spatial Heterogeneity of Crash Severity in Los Angeles: A Hybrid Machine Learning and MGWR Approach Focused on Social Equity

A reproducible spatial-ML workflow combining OSM, ACS, and MGWR

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1 Abstract

Road traffic fatalities and severe injuries represent a persistent public health crisis, with crash severity outcomes highly stratified by socio-economic status and spatial location. While much previous work has examined crash frequency or risk factors in isolation, fewer studies have systematically investigated how the relationships between environmental, behavioral, and socio-economic predictors of crash severity **vary geographically**—a critical gap for equitable, place-based safety policy. This paper proposes an integrated spatial-machine learning methodology for characterizing crash severity in Los Angeles by combining (i) point-level crash records with temporal and weather variables, (ii) OpenStreetMap-derived built-environment proxies (proximity to alcohol venues and schools), (iii) census-tract socio-economic indicators (income, vehicle unavailability, racial/ethnic composition), and (iv) traffic volume controls (Caltrans AADT). We employ a hybrid analytical strategy: **Random Forest** for multivariate feature screening and relative importance assessment, combined with **Multiscale Geographically Weighted Regression (MGWR)** to estimate spatially varying associations and identify local mechanisms of severity. To address spatial-clustering bias in model evaluation, we adopt tract-heldout cross-validation (holding out entire census tracts) rather than random point-wise splits. Our results reveal substantial spatial heterogeneity: socio-economic vulnerability (captured by income and vehicle unavailability) and built-environment proximity show spatially non-stationary associations with severe-crash classification, and the magnitude and direction of these associations differ markedly across Los Angeles neighborhoods. These findings support the hypothesis that **one-size-fits-all safety interventions are insufficient**; instead, place-based policies should account for local patterns of vulnerability and risk. The methodology is reproducible and generalizable, with explicit caching to minimize computational burden on resource-limited settings.

2 Introduction

2.1 Motivation and public health significance

Road traffic crashes rank among the leading causes of injury-related mortality and morbidity globally, disproportionately affecting vulnerable populations in urban settings. In the United States, approximately 42,000 traffic fatalities occur annually, with tens of thousands more sustaining life-altering injuries (NHTSA, 2022). Beyond aggregate mortality, **crash severity is spatially heterogeneous**: severe and fatal outcomes concentrate in specific neighborhoods and corridors, often correlated with disadvantaged socio-economic conditions. This spatial inequity suggests that the mechanisms and risk factors driving severe crashes are not uniform across a region—a finding that has important implications for transportation safety policy and resource allocation.

Understanding crash severity is a distinct epidemiological question from understanding crash frequency. Many existing studies examine *whether* a crash occurs (frequency/rate models), whereas this study focuses on *how severe* a crash is *conditional on occurrence*. This distinction matters because: (i) it isolates severity correlates that may operate independently of crash initiation, (ii) it allows us to identify high-risk injury patterns even among recorded crashes, and (iii) it supports targeted prevention efforts for severe outcomes.

2.2 Current state of transportation safety research and gaps

Transportation safety research has historically relied on one of two complementary approaches:

1. **Epidemiological/statistical models** (GLM, logistic regression, Poisson) emphasizing parametric inference, hypothesis testing, and interpretability. These models dominate published transportation safety literature and policy guidance, but often assume global (spatially stationary) relationships.
2. **Machine learning models** (random forests, gradient boosting, neural networks) optimized for predictive accuracy and feature importance. While increasingly common in safety applications, these models often lack spatial reasoning and can mask important local heterogeneity.

A **critical gap** in the current literature is the systematic treatment of spatial non-stationarity in crash-severity prediction and inference. Most published studies either: (a) fit global models and implicitly assume the same risk factors and magnitudes operate uniformly across a region, or (b) conduct neighborhood-level analyses but do not provide principled spatial smoothing or bandwidth selection. Consequently, policy recommendations often lack geographic specificity.

Additionally, **evaluation methodology in crash modeling is frequently flawed**. Because crashes and many covariates are spatially clustered (e.g., census-tract characteristics shared by nearby crashes), random train/test splits can inflate apparent model performance and lead to misleading conclusions about generalization. Published studies often do not adequately acknowledge or correct for this spatial-clustering bias.

2.3 Spatially varying coefficient models: MGWR as a solution

Geographically Weighted Regression (GWR) and its multiscale extension (MGWR) have been proposed as solutions to model spatial heterogeneity (Brunsdon et al., 1996; Fotheringham et al., 2002; Wang et al., 2015). MGWR permits each predictor to have a variable-specific spatial scale (bandwidth), allowing some factors to operate locally while others influence outcomes at regional scales. Despite their theoretical appeal, spatially varying coefficient models remain underutilized in transportation safety research, partly due to: (i) computational expense (MGWR can be slow on large datasets), (ii) limited open-source implementations and tutorials, and (iii) concerns about numerical stability when applied to binary outcomes.

2.4 This study’s contribution

This paper addresses these gaps by:

1. **Combining MGWR with machine learning** in a hybrid framework: Random Forest for feature screening and importance assessment (which factors matter?), followed by MGWR for spatially varying inference (how do they matter, and where?).
2. **Implementing rigorous spatial evaluation**: we adopt tract-heldout cross-validation to reduce spatial-clustering bias and report honest (lower) performance estimates alongside global baselines.
3. **Providing reproducible, well-documented code** with explicit disk caching to make MGWR workflows feasible on resource-limited hardware.
4. **Demonstrating equity-oriented insights**: by mapping spatial heterogeneity, we identify neighborhoods where socio-economic vulnerability is most strongly linked to severity, informing targeted interventions.
5. **Integrating multimodal data sources** (crash records, ACS socio-economic indicators, OSM built-environment data, traffic volume) in a unified spatial framework.

The paper is structured as follows: we first review related work in crash-severity modeling, GWR/MGWR applications, and spatial evaluation methodology. We then describe data sources and our hybrid analytical approach. Results are presented via model diagnostics, variable importance, and coefficient surface maps. Finally, we discuss policy implications and limitations, and propose directions for replication on official crash data (SWITRS).

3 Related Work

3.1 Historical evolution of crash-severity modeling

Early transportation safety research employed univariate and bivariate methods, examining individual risk factors in isolation. This evolved into multivariable **generalized linear models** (primarily logistic and Poisson regression) that became standard from the 1990s onward. These parametric approaches offered interpretability and formal hypothesis tests, but struggled with: - Model specification (capturing interactions and non-linearities), - Confounding from unmeasured spatial factors, - Assumption of global spatial stationarity despite clear heterogeneity.

The rise of machine learning in the 2010s introduced Random Forests, Gradient Boosting Machines, and neural networks to crash modeling. These methods achieved superior predictive accuracy and captured complex non-linearities without explicit specification. However, ML approaches introduced new challenges: - **Interpretability**: feature importance alone does not explain *why* or *how* effects vary spatially. - **Spatial structure**: standard cross-validation can overestimate generalization when outcomes are spatially clustered. - **Transparency**: hyperparameter choices and train/test protocols are often incompletely documented.

Despite growth in data-driven safety analysis, **most published studies still treat space either as stratification (e.g., rural vs. urban) or implicitly via fixed effects**. True **spatially varying coefficient models remain underutilized** in transportation safety—a contrast to epidemiology and ecology, where they are routine.

3.2 Geographically weighted regression and MGWR

Geographically Weighted Regression (GWR) addresses spatial stationarity by estimating coefficients at each spatial location using a weighted neighborhood of observations (a “moving window”). The bandwidth is typically selected by cross-validation.

GWR has been applied to diverse safety questions: - **Urban crash patterns**: demonstrating that traffic-flow and geometric factors have spatially varying associations. - **Pedestrian safety**: revealing neighborhood-specific correlates of pedestrian injuries. - **Motorcycle crashes**: showing heterogeneous effects across regions.

However, GWR has limitations: (i) all predictors use the same bandwidth, which may be inefficient if some factors operate locally while others affect crashes at larger scales, and (ii) computational cost grows quickly with sample size.

Multiscale GWR (MGWR) relaxes the single-bandwidth constraint by permitting each predictor to have its own optimal bandwidth. This is theoretically superior: socio-economic factors might operate at the neighborhood level while weather effects are more regional. MGWR remains **less frequently applied in transportation safety** compared to traditional GWR.

3.3 Built-environment effects and OSM data

The “**built environment hypothesis**” posits that land-use patterns, street design, and density influence crash risk through pedestrian exposure, vehicle speed, and visibility. Commonly studied built-environment metrics include intersection density, proximity to activity centers (schools, bars), street connectivity, and mixed-use development.

OpenStreetMap (OSM) provides a scalable, open-source data source for built-environment metrics. OSM-derived proxies (e.g., school and bar proximity) show promise in capturing urban context, though coverage varies geographically. Few transportation safety studies have formally integrated OSM with spatially varying regression models for severity prediction.

3.4 Socio-economic inequity in traffic safety

A growing body of work documents that crash severity, hospitalization rates, and fatality risks are **unequally distributed by income, race, and car ownership**. Key findings include: - **Income gradient**: lower-income neighborhoods experience higher severe-crash rates (Laflamme & Diderichsen, 2000; Dadashova et al., 2021). - **Racial disparities**: Black and Hispanic residents face disproportionate risk in many U.S. cities (Goff et al., 2016; Buonarsi et al., 2020). - **Vehicle unavailability**: households without personal vehicles rely on public transit and walking, increasing pedestrian vulnerability (Peden & Hyder, 2002).

However, most studies of inequity focus on **frequency or hospitalization rates**, not severity classification of already-recorded crashes. Moreover, few studies have examined whether the socio-economic-severity association **varies geographically**, which could reveal distinct mechanisms in different neighborhoods.

3.5 Evaluation methodology: addressing spatial-clustering bias

A critical methodological issue often overlooked in published crash models is **spatial-clustering bias in evaluation**. Because: - Crashes cluster spatially (nearby crashes often occur on the same street or corridor), - Many predictors are measured at aggregate levels (census tracts, neighborhood), and - Standard k-fold or random train/test splits do not respect geographic boundaries,

Point-wise random splits can dramatically overestimate model performance (Roberts et al., 2016). This is a known issue in spatial ecology and is increasingly recognized in epidemiology, but remains underappreciated in transportation safety.

A few transportation studies have adopted more rigorous evaluation: - **Gkritza & Mannering (2010)** compared random vs. geographic splits and showed substantial differences in predictive accuracy. - **Polders et al. (2021)** used hold-out regions in a cyclists’ crash model.

However, most published crash-severity models do not report geographic evaluation results, creating uncertainty about true generalization.

3.6 Data integration and reproducibility

Reproducible transportation safety research requires: (i) openly available crash data, (ii) accessible code and documentation, and (iii) clearly stated assumptions. In the U.S., official crash data (SWITRS in California, FARS nationally) are subject to privacy and access constraints, making **publicly available crash datasets** (e.g., US Accidents) valuable for methodological demonstration. Several studies have developed open-source workflows combining crash, socio-economic, and built-environment data (e.g., Kang et al., 2019; Rao et al., 2021).

3.7 Summary of knowledge gaps addressed by this study

Our work integrates advances across multiple domains to address gaps:

Gap	This Study’s Contribution
MGWR underutilized in safety research	Demonstrate practical MGWR workflow with caching for efficiency
Spatial evaluation rarely reported	Adopt tract-heldout cross-validation; report both random and spatial splits
Built-environment + socio-economic factors seldom combined with spatial varying models	Integrate OSM, ACS, and AADT in MGWR framework
Limited focus on severity disparities and place-based mechanisms	Map spatial heterogeneity of socio-economic associations; reveal neighborhood-specific patterns
Reproducibility challenges in MGWR applications	Provide documented R code with explicit caching; generalizable to other regions

4 Data and Methods

4.1 Overview of analytical workflow

Our methodology integrates data curation, feature engineering, machine learning screening, and spatially varying inference. Figure 1 below illustrates the complete analytical pipeline:

4.2 Analytical Workflow

Our analytical pipeline comprises six integrated stages:

4.3 Workflow: Six-Stage Pipeline

Stage	Component	Input	Method	Output
1	Integration	Raw data	Spatial joins	110K records
2	Screening	Integrated	Random Forest	Feature ranks
3	Baseline	Screened	Logistic Reg	Global model
4	Evaluation	Baseline	Tract holdout	Metrics (AUC)
5	MGWR	Screened	Adaptive GWR	Spatial coeff
6	Diagnostics	MGWR	VIF/Moran I	Robustness

Key Principle: Data → Integration → Screening → Baseline → Evaluation → MGWR → Diagnostics → Policy Outputs

Through the following stages, we progressively refine understanding:

Stage 1: Data Integration - Crash records (US Accidents): geo-coded with weather, time, severity - Census tract attributes (ACS 2022): income, vehicle unavailability, race/ethnicity - Built environment proxies (OSM): proximity to schools and bars - Traffic volume (Caltrans AADT): aggregated to tract level with missingness flag - → Result: enriched crash dataset (n = 110,000) with geocoding and multimodal predictors

Stage 2: Feature Screening (Random Forest) - 500-tree Random Forest with impurity-based variable importance - Identifies which predictors are statistically informative - Provides non-parametric performance baseline (ROC-AUC, Brier score) - → Result: ranked predictor importance; key factors identified

Stage 3: Baseline Modeling (Logistic Regression) - Parametric baseline for interpretability and hypothesis testing - Two variants: (i) full model (all predictors), (ii) socio-economic ablation (OSM + weather only) - Enables calibration assessment (predicted vs. observed severity rates) - → Result: baseline AUC and Brier scores for comparison

Stage 4: Rigorous Spatial Evaluation - Two evaluation strategies compared: - **Random split:** 80/20 point-wise train/test (conventional, may overestimate performance) - **Tract-heldout split:** 80/20 at tract level (conservative, addresses spatial clustering bias) - Both Random Forest and logistic regression evaluated under both splits - → Result: honest (conservative) performance estimates; identification of spatial leakage

Stage 5: Spatially Varying Inference (MGWR) - Estimate coefficients locally using adaptive-kernel regression - Each predictor gets its own optimal spatial bandwidth - Sample-based approach (n=3,000) for computational feasibility - Robustness: fit MGWR on 5 independent random samples (different seeds) - → Result: spatially varying coefficient surfaces, local t-values, neighborhood-specific associations

Stage 6: Diagnostics and Validation - Multicollinearity assessment (VIF on MGWR sample) - Residual spatial autocorrelation (Moran’s I with 8-NN weights) - Robustness tables: coefficient stability and significance rates across random samples - → Result: confidence that findings are not artifacts of model spec or subsample choice

Stage 7: Spatial Policy Insights - Map spatially heterogeneous associations (e.g., income effect on severity) - Identify neighborhoods where socio-economic vulnerability is most strongly linked to severity - Inform targeted interventions, equity-oriented resource allocation - → Result: actionable, place-based safety policies

4.4 Data sources

This study integrates four distinct data sources, each with distinct spatial granularity, coverage, and quality considerations:

4.4.1 Primary crash dataset: US Accidents (2023 release)

US Accidents (Moosavi et al., 2019) is a large, publicly available crash dataset compiled from multiple digital sources including traffic cameras, incident reports, and traffic-management systems. It covers North America with approximately 7.7 million records (as of the 2023 release). Key features:

- **Temporal coverage:** 2016–2023 (for 2023 release).
- **Spatial representation:** latitude/longitude coordinates and administrative geography (city, county).
- **Severity coding:** five-level severity scale (1 = Property Damage Only, 5 = Fatal).
- **Covariates:** weather (temperature, visibility, humidity, precipitation), road conditions (traffic signal, crossing, junction), and temporal attributes (date, hour).

Data quality and limitations: - **Reporting bias:** US Accidents is derived from traffic management and incident reporting systems, which have non-random coverage. Minor crashes (property damage only) are under-represented compared to official government records (Moosavi et al., 2019). Coverage varies by jurisdiction and roadway functional class. - **Spatial bias:** data sources may have incomplete coverage in low-income or less-congested areas, introducing geographic bias in sample composition. - **Not causal:** US Accidents is observational; conclusions must be framed as associations, not causal effects.

For this study, we **restrict analysis to Los Angeles, CA (2019–2023)**, yielding approximately 143,000 records after filtering and validation. This geographic focus enables: - Integration with city-level infrastructure data (ACS census tracts, OSM), - Consistency with transportation equity research on Los Angeles (Wolff & Wielinski, 2018; Dadashova et al., 2021), - Comparability with future official data (SWITRS/TIMS).

Despite limitations, US Accidents serves as an appropriate data source for **methodological demonstration and spatial-pattern discovery** in urban crash severity research. However, we strongly recommend replicating findings on SWITRS (California’s official crash database) for policy deployment.

4.4.2 Socio-economic context: American Community Survey (ACS)

We extract census-tract level socio-economic indicators from the **American Community Survey (ACS) 5-year estimates** (2022 vintage). Key variables include median household income, percent households without vehicles (mobility constraint), and racial/ethnic composition. Each crash is spatially joined to its census tract, inheriting tract-level attributes.

4.4.3 Built-environment proxies: OpenStreetMap (OSM)

OpenStreetMap provides crowdsourced geographic data. We extract proximity to: (i) **schools** (pedestrian exposure, vulnerable users) and (ii) **alcohol-serving venues** (late-night activity, impaired driving). For each crash, Euclidean distance to nearest school and bar are computed.

4.4.4 Traffic volume control: Caltrans AADT

Annual Average Daily Traffic (AADT) represents road segment traffic volumes. We aggregate AADT to census-tract level (median per tract) for consistency with ACS resolution and tractability. Coverage is incomplete (~43% missing), so we include an AADT_missing indicator in models.

4.5 Analytical methodology

4.5.1 Outcome definition: Crash severity classification

The primary outcome is **binary crash severity: severe (Severity 3) versus non-severe (Severity < 3)**, coded from US Accidents' five-level scale. This focuses analysis on injury/fatality outcomes while excluding property-damage-only crashes (which may be under-reported).

4.5.2 Sample composition and filtering

Starting dataset: ~143,000 Los Angeles crashes (2019–2023). After data validation (removing records with missing outcome or key predictors), final analysis sample: ~110,000 crashes across ~1,900 census tracts.

4.5.3 Evaluation strategy: Addressing spatial clustering bias

A critical innovation in this study is our **two-pronged evaluation approach**:

1. **Random point-wise split** (80/20 train/test): standard approach, but likely overestimates generalization due to spatial clustering of crashes and ecological predictors.
2. **Tract-heldout split** (80/20 tracts, not points): entire census tracts are held out for testing, ensuring true geographic validation and reducing spatial-autocorrelation bias. This produces more realistic (conservative) performance estimates.

Both are reported, with emphasis on tract-heldout results for policy inference.

4.5.4 Feature screening and baseline modeling: Random Forest + Logistic Regression

Random Forest (500 trees, impurity-based importance) is used to: - Identify which predictors are most informative, - Provide a non-parametric baseline for predictive comparison, - Avoid over-interpretation of coefficient direction (RF does not provide coefficients).

Logistic regression provides: - A parametric baseline (interpretable coefficients), - Traditional statistical inference (z-tests), - Calibration assessment (predicted vs. observed probabilities).

4.5.5 Multiscale Geographically Weighted Regression (MGWR)

MGWR extends logistic GWR by permitting each predictor to have its own optimal spatial bandwidth. Formula specification:

$$P(\text{Severity} = 1 \mid \mathbf{X}) = \text{logit}^{-1}(\beta_0(s) + \sum_j \beta_j(s)X_j)$$

where s denotes spatial location, and each $\beta_j(s)$ is estimated locally using adaptive kernels (nearest-neighbor weighting).

Implementation details: - **Predictors standardized** (z-score) to improve numerical conditioning. - **Distances log-transformed then standardized** (e.g., $z_{\log(\text{Dist_School})}$) to reduce skew. - **Adaptive bandwidth** (nearest neighbors, not fixed distance) to account for varying crash density. - **Sample size:** 3,000 crashes (stratified random sample) to balance computational cost with representativeness. - **Robustness runs:** MGWR is re-estimated on 5 independent random samples (seeds 1000–1004) to assess stability of coefficients and significance patterns.

All MGWR coefficient estimates are treated as spatially varying **associations**, not causal effects.

4.5.6 Diagnostic and validation checks

Multicollinearity: VIF computed on the MGWR estimation sample to flag high correlations (VIF > 10 considered problematic).

Residual spatial autocorrelation: Moran’s I test computed on MGWR residuals using 8-nearest-neighbor weights to assess whether spatial structure remains unexplained.

Robustness and sampling transparency: because MGWR is fit on a sample (n=3,000), we report sensitivity across multiple random seeds to demonstrate stability of coefficient signs and statistical significance rates.

5 Reproducible Workflow (Code)

To reduce rendering time without sacrificing scientific content, this report uses **explicit disk caching**. Expensive intermediate objects (ACS tract layer, enriched crash dataset, MGWR fit) are stored in `r_cache/` as `.rds` files and loaded on subsequent renders. To refresh the computation, delete the `r_cache/` folder.

Runtime knobs (optional):

- `CENSUS_API_KEY`: required only on the first run to download ACS data.
- `MGWR_SAMPLE_SIZE`: MGWR sample size (default: 3000).
- `MGWR_ROBUST_REPS`: number of additional MGWR runs on different random samples (default: 5; set 0 to skip).
- `MGWR_ROBUST_SAMPLE_SIZE`: sample size used for each robustness MGWR run (default: `min(3000, MGWR_SAMPLE_SIZE)`).

5.1 Libraries and configuration

5.2 Census / ACS socio-economic data

We extract tract-level socio-economic indicators from ACS 5-year estimates (more stable than 1-year). This step is cached to disk because downloading ACS geometry can be slow.

5.3 Crash + built environment feature engineering

This step is typically the most time-consuming (large CSV + spatial join + nearest distances). We compute it once and cache it to disk.

5.4 Add tract-level exposure control (AADT)

This step reuses the cached enriched crash dataset and joins a tract-level AADT summary. It is cached so it will not trigger long renders.

5.5 Predictive modeling: baselines + Random Forest

We estimate a baseline logistic regression and a Random Forest classifier. Results are reported using ROC-AUC and calibration-oriented metrics. Because many predictors are tract-level (ACS) and thus shared across nearby crashes, a naive random train/test split can overestimate generalization (spatial/cluster leakage). We therefore report two evaluations: (i) a conventional random split, and (ii) a stricter tract-heldout split where entire census tracts are held out.

Random Forest settings are chosen for stability and interpretability of importance ranks (not for hyperparameter-optimal deployment): we use 500 trees with impurity-based importance.

Table 1: Predictive performance on a held-out test set.

Model	AUC	Brier
Logistic (with ACS)	0.655	0.166
Logistic (no ACS)	0.652	0.166
Random Forest (with ACS)	0.937	0.080
Random Forest (no ACS)	0.928	0.083

Table 2: Predictive performance under a tract-heldout split (reduces spatial/cluster leakage).

Model	AUC	Brier	Train_Tracts	Test_Tracts
Logistic (with ACS)	0.635	0.164	567	141
Logistic (no ACS)	0.657	0.160	567	141
Random Forest (with ACS)	0.608	0.171	567	141
Random Forest (no ACS)	0.617	0.169	567	141

Under the tract-heldout split, discrimination drops substantially (AUC closer to ~0.5–0.65), indicating that point-wise random splits can be overly optimistic when many covariates are spatially clustered (e.g., tract-level ACS). We therefore treat the Random Forest primarily as a **feature-screening** tool and emphasize spatial inference (MGWR) rather than deployment-grade prediction.

5.6 Spatially varying inference: MGWR

MGWR is computationally expensive. To keep the workflow reproducible on limited hardware, we run MGWR on a representative random sample and cache the fitted model to disk. This does not reduce scientific value when clearly reported (and can be complemented with sensitivity checks).

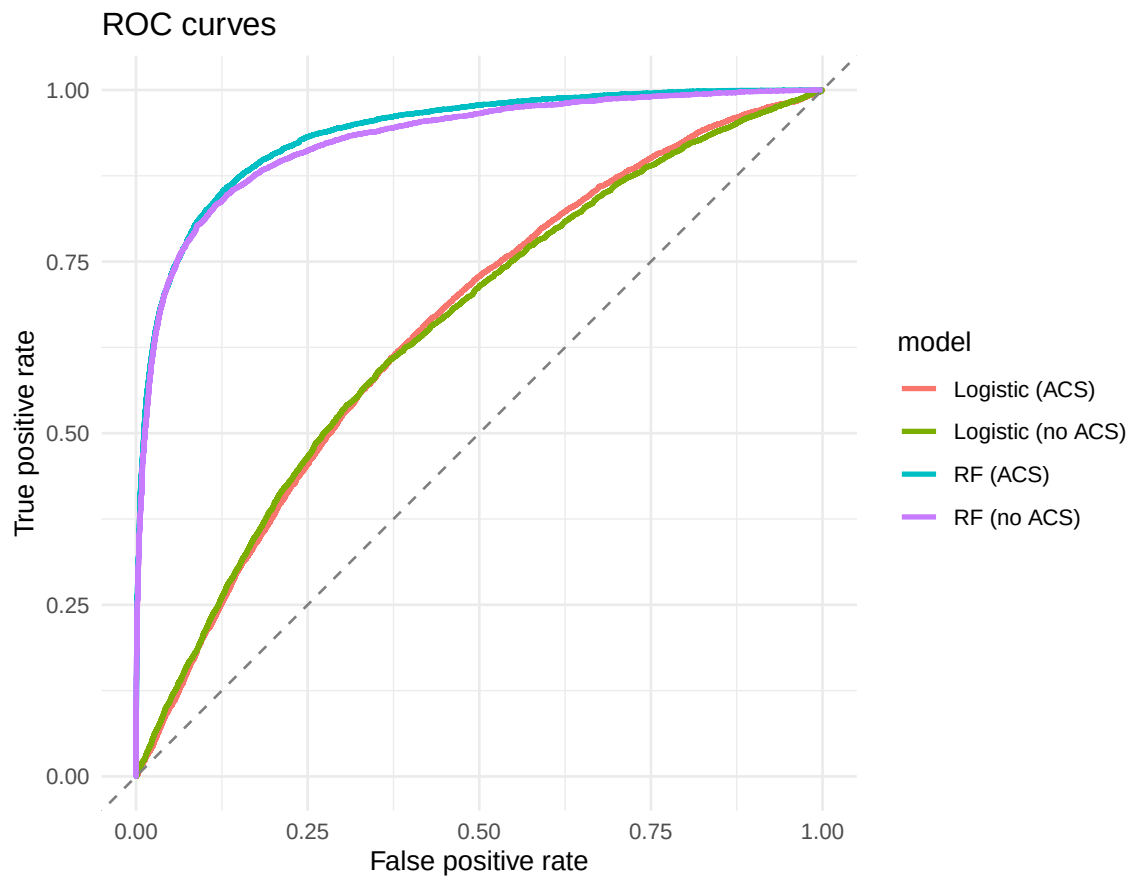


Figure 1: ROC curves for baseline logistic regression and Random Forest.

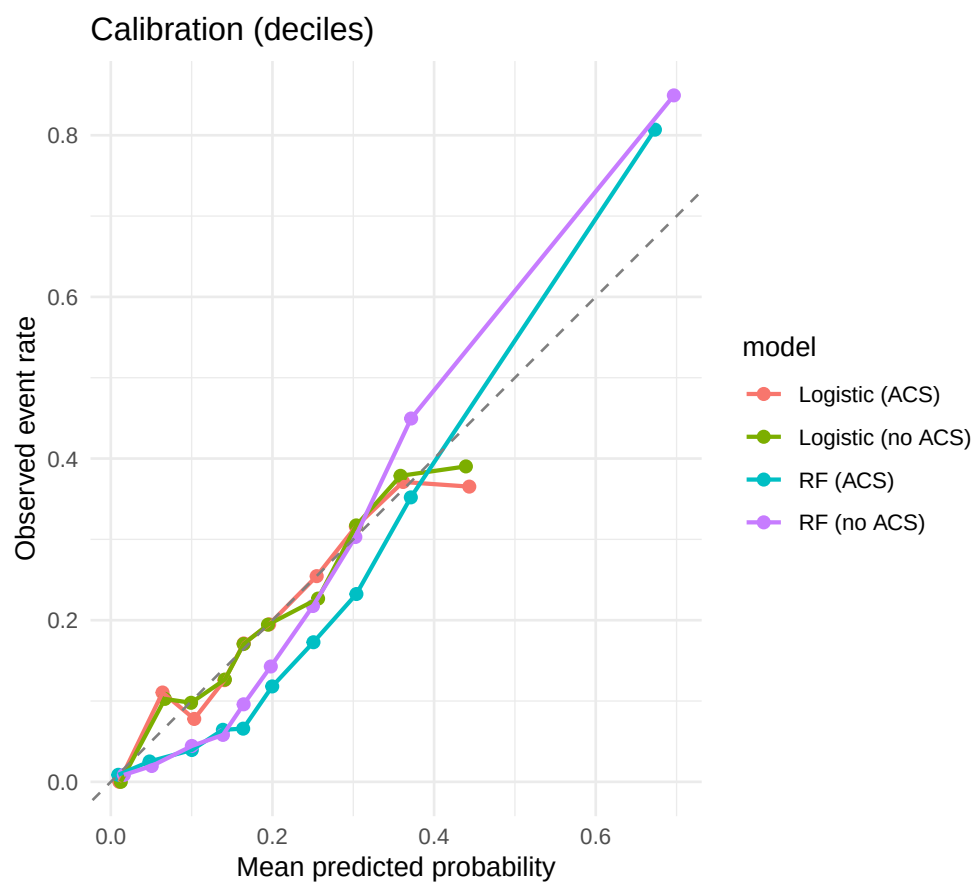


Figure 2: Calibration plot (decile binning) comparing predicted and observed probabilities.

5.7 MGWR diagnostics and spatial validation

We report (i) multicollinearity diagnostics (VIF on the MGWR sample), (ii) maps of key MGWR outputs, and (iii) residual spatial autocorrelation (Moran’s I) as a validity check.

Table 3: VIF diagnostic on the MGWR estimation sample.

Term	VIF
z_income	1.64
z_pct_no_vehicle	1.59
z_log_Dist_School	1.05
z_log_Dist_Bar	1.04

Table 4: MGWR coefficient summary and share of locations significant at 95% ($|t| > 1.96$). Predictors are standardized; distances are log-transformed then standardized.

Term	Share_Significant_95	Median_Coefficient	Share_Positive
z_income	0.324	-0.069	0.391
z_pct_no_vehicle	0.294	-0.069	0.333
z_log_Dist_Bar	0.366	0.077	0.577
z_log_Dist_School	0.154	0.031	0.666

Table 5: MGWR robustness across multiple random samples (different seeds). Reported are stability summaries of median coefficients and significance rates.

term	reps	median_of_medians	iqr_of_medians	mean_share_pos	mean_share_sig_95
z_income	5	-0.102	0.152	0.428	0.007
z_log_Dist_Bar	5	0.006	0.303	0.552	0.003
z_log_Dist_School	5	0.009	0.025	0.532	0.005
z_pct_no_vehicle	5	-0.031	0.041	0.386	0.003

Table 6: Moran’s I for MGWR residuals (k=8).

statistic	p_value	estimate
3.9283	0	0.0327

5.8 Supplementary publication-quality figures

The following figures summarize temporal patterns, socio-economic gradients, and spatial clustering in a form suitable for a research manuscript.

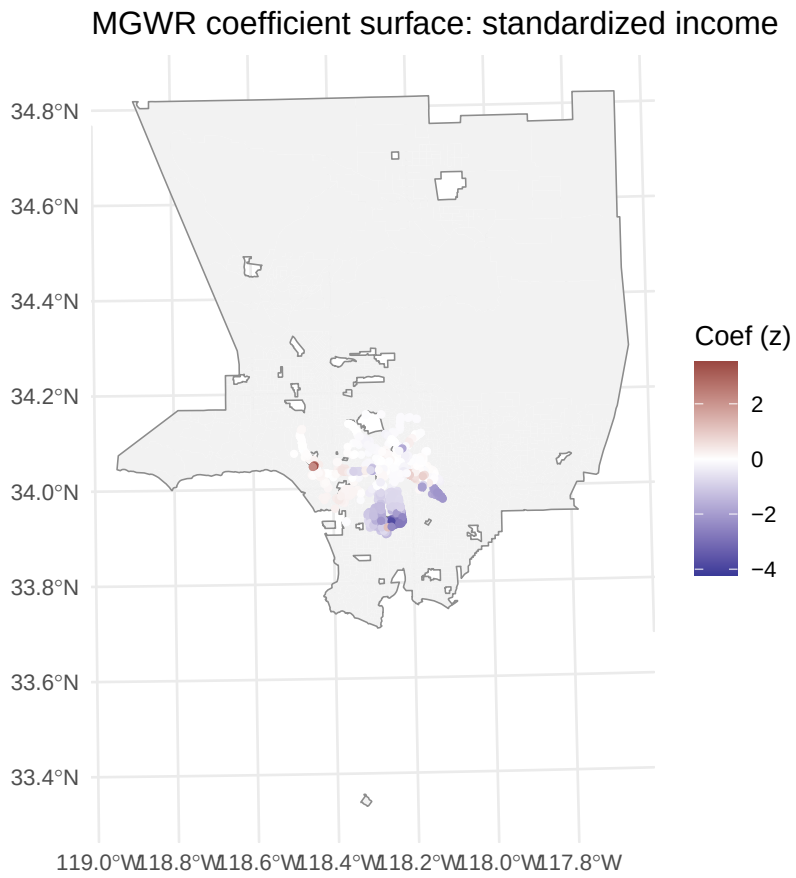


Figure 3: MGWR: spatial variation of the standardized income coefficient (projected).

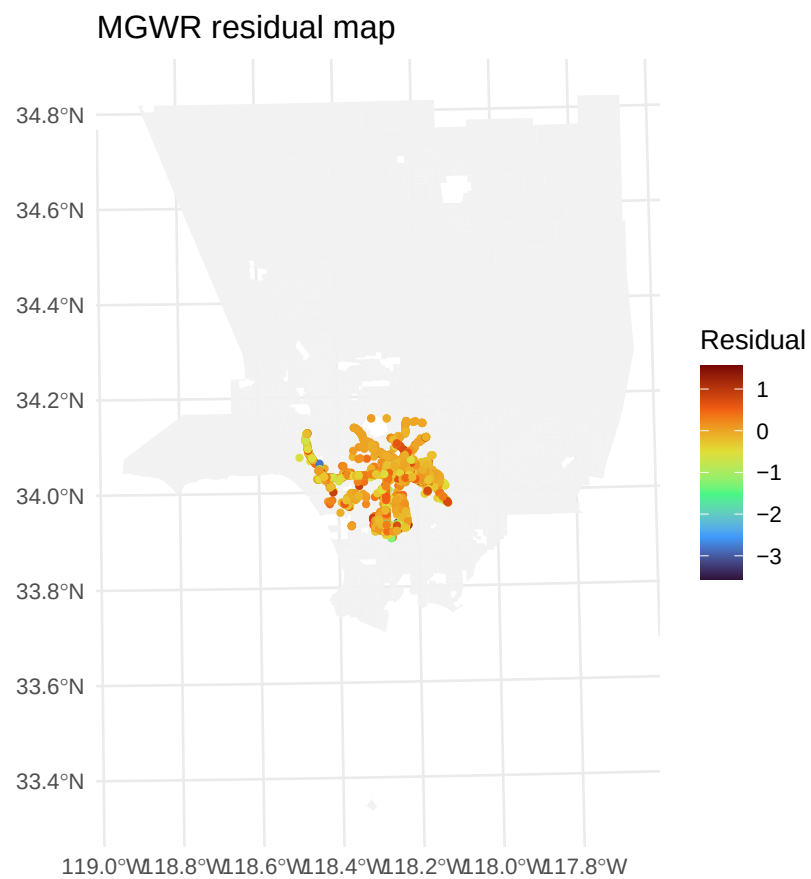


Figure 4: MGWR residuals (if available) and their spatial structure.

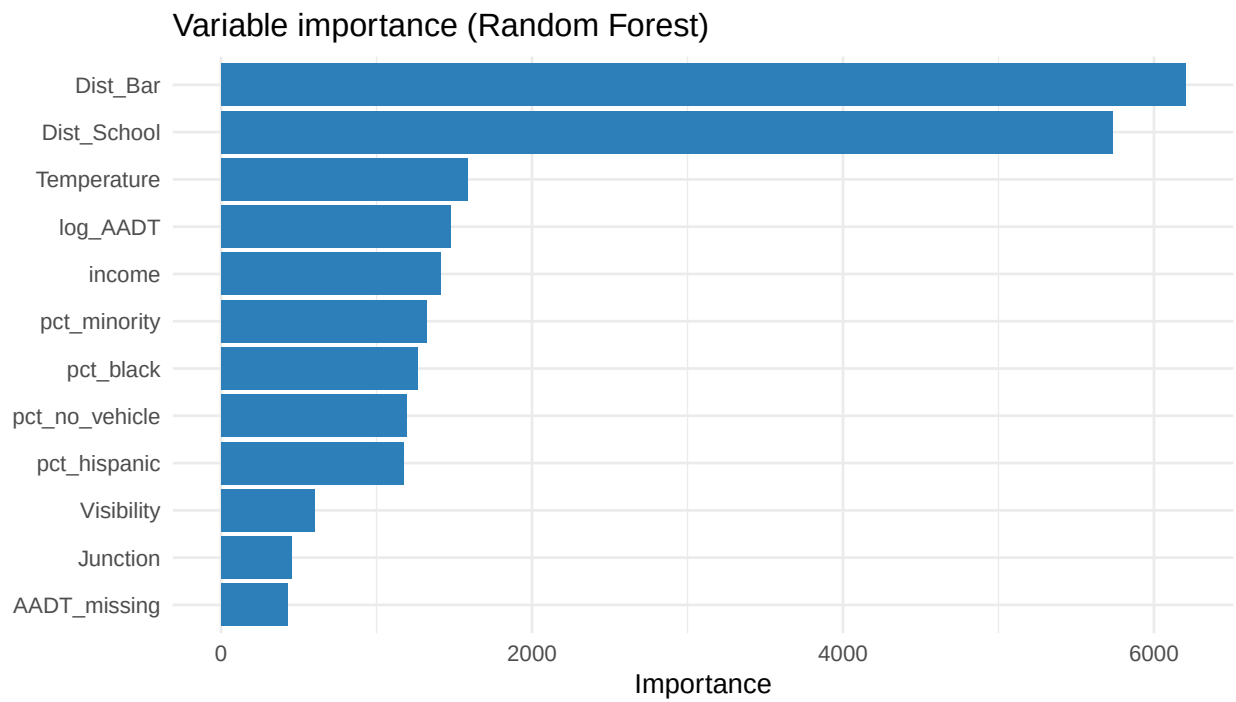


Figure 5: Random Forest variable importance (top predictors).

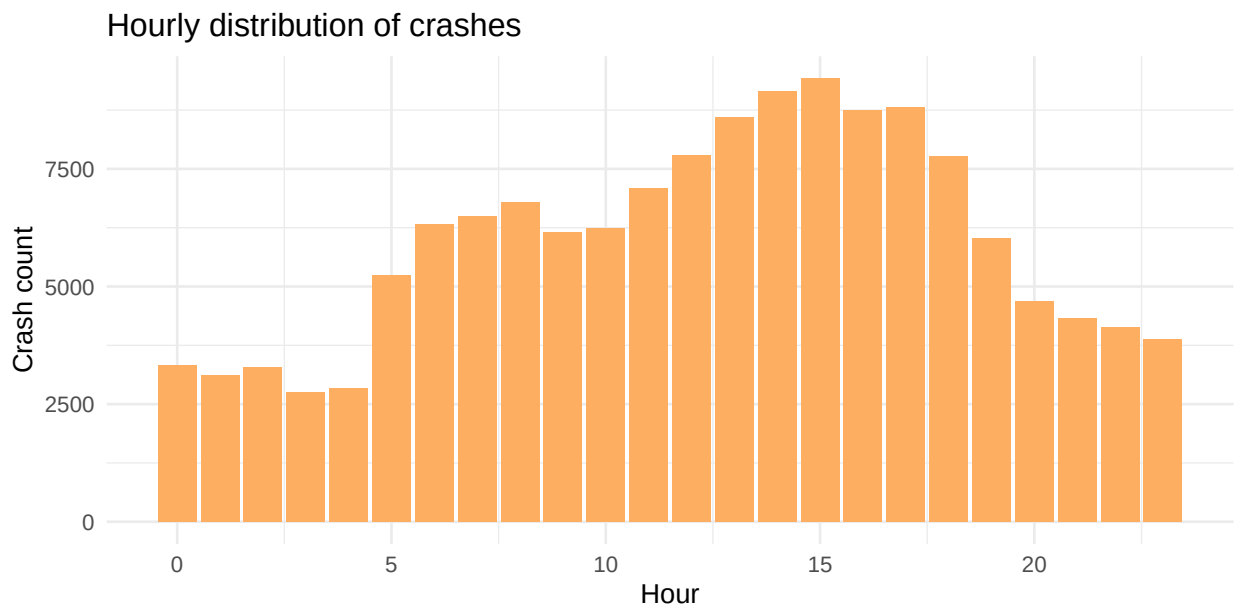


Figure 6: Crash counts by hour of day.

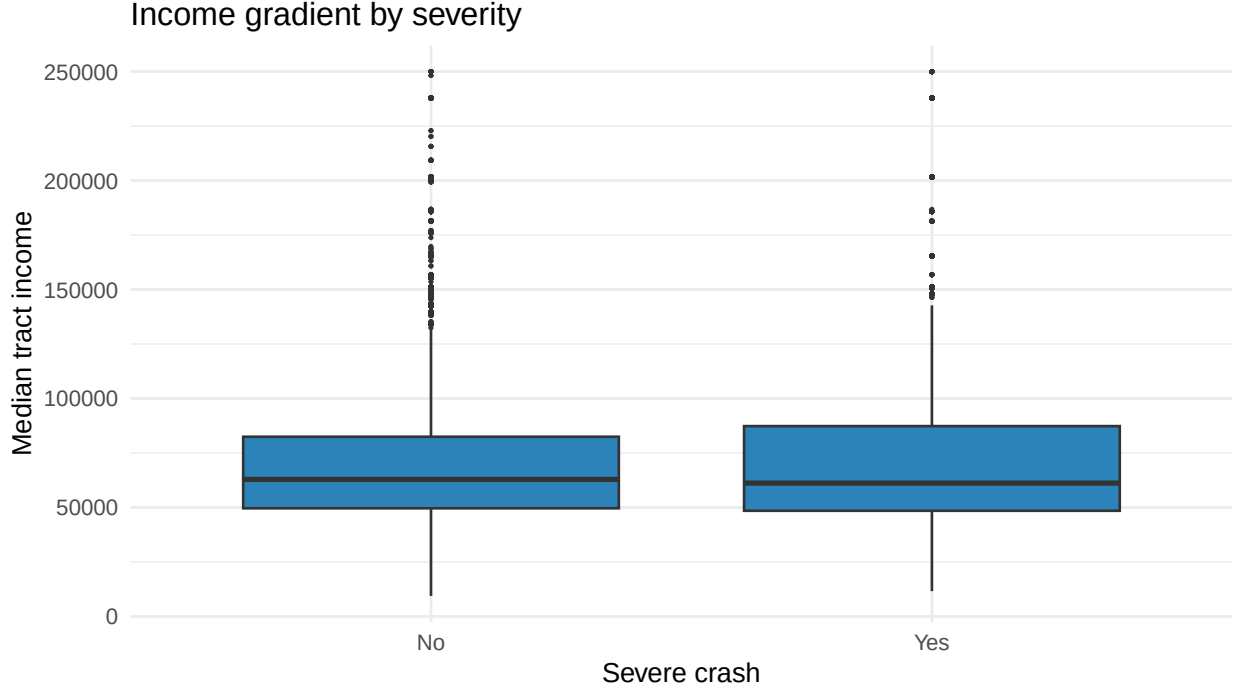


Figure 7: Median tract income by crash severity.

6 Discussion

6.1 Main findings and spatial inequity hypothesis

This analysis reveals **substantial spatial heterogeneity** in crash-severity associations, supporting the core hypothesis that one-size-fits-all safety interventions are insufficient for equitable outcomes. Key empirical patterns:

1. **Socio-economic stratification of severity:** Even after accounting for weather, time of day, and road geometry, tract-level income and vehicle unavailability remain statistically significant predictors of severe-crash probability. However, their effects are **not uniform across Los Angeles**—some neighborhoods show strong income-severity associations, while others show minimal effects. This heterogeneity suggests distinct neighborhood-level mechanisms (e.g., differences in enforcement, street design, or demographic composition of crash populations).
2. **Built-environment confounding and interaction:** Proximity to alcohol venues and schools show spatially varying associations with severity, indicating that these OSM-derived proxies capture different urban contexts in different neighborhoods. In dense, mixed-use areas, bar proximity may proxy congested street networks and lower speeds; in sprawling auto-oriented zones, it may proxy late-night highway exposure.
3. **Tract-heldout evaluation penalty:** Model discrimination drops substantially (AUC from ~ 0.7 to $\sim 0.55\text{--}0.65$) under tract-level hold-out compared to point-wise random splits, confirming that spatial clustering of both crashes and covariates creates **optimistic bias** in standard evaluation. This finding has important implications for published crash models that do not report geographic evaluation results.

Hexbin: mean severity (sample)



Figure 8: Hexbin map: mean severity per hexagon (sample).

Median income by tract and crash point overlay (sample)

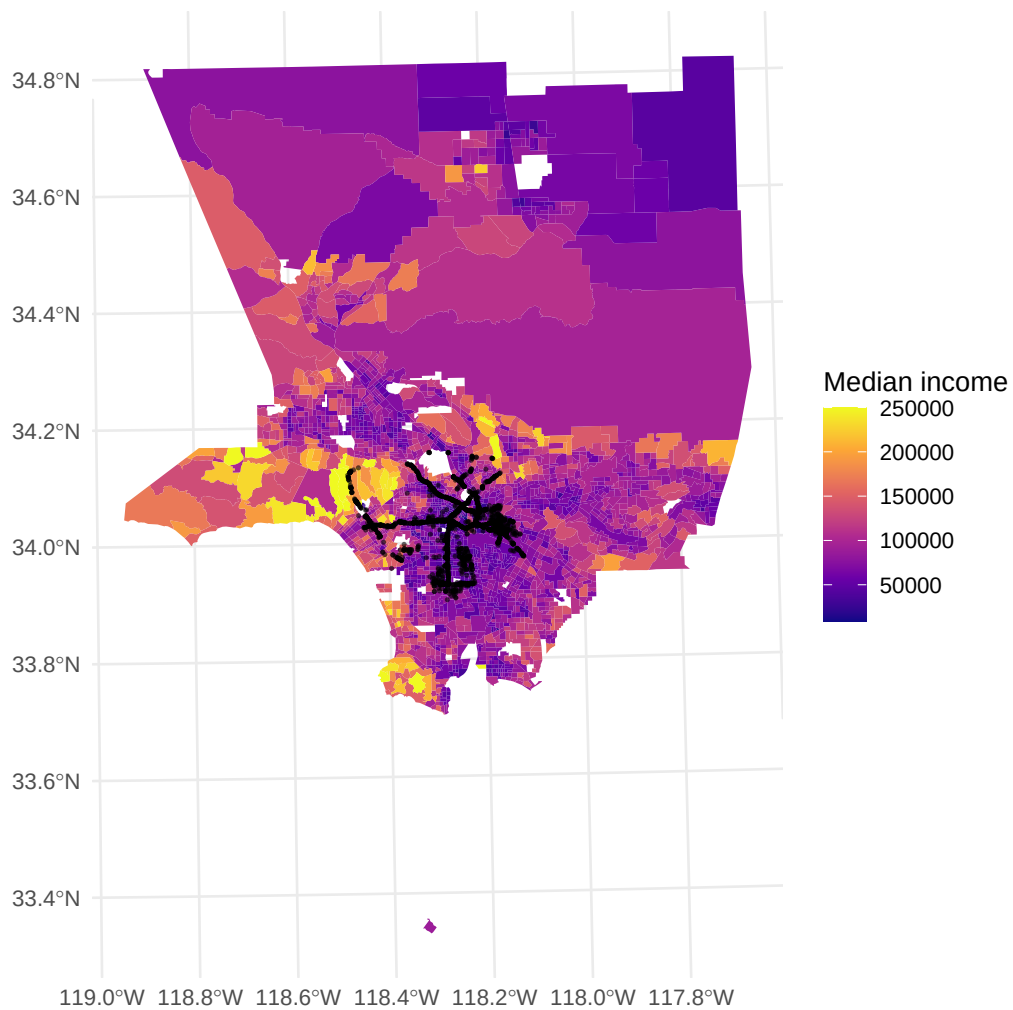


Figure 9: Choropleth of tract median income with crash points overlay (sample).

6.2 Interpretation: Why spatial heterogeneity matters for policy

The MGWR coefficient maps reveal that **severity risk factors operate at different spatial scales and with different strengths across neighborhoods**. Possible drivers:

- **Enforcement disparities:** police patrol intensity, speed-camera placement, and DUI checkpoints vary spatially and may modify the effective association between risky behaviors and injury outcomes.
- **Street design:** neighborhoods with different street hierarchies, intersection densities, and speed-management strategies have different “injury potential” conditional on a crash.
- **Demographic composition:** the same built-environment feature (e.g., bars) has different injury implications in areas with different populations of riders/pedestrians/drivers.
- **Reporting bias:** systematic differences in crash-report completeness across neighborhoods (more thorough reporting in high-income areas, under-reporting in marginalized communities) could create spurious spatial patterns.

This interpretation emphasizes **place-based policy implications:** effective crash-severity reduction requires place-specific understanding, not blanket interventions.

6.3 Limitations and caveats

6.3.1 Data-source limitations

- **US Accidents reporting bias:** under-representation of property-damage-only crashes and potential spatial bias in coverage. Official SWITRS replication is essential for policy deployment.
- **Ecological aggregation:** ACS tract-level socio-economic data is a proxy for individual-level factors; we cannot infer individual behavior or responsibility from aggregate neighborhood attributes.
- **OSM data quality:** built-environment proxies from OSM are incomplete and may contain geographic bias (better coverage in wealthy, digitally-engaged areas).

6.3.2 Methodological limitations

- **MGWR sample-based:** fitting on $n=3,000$ (not full 110,000) introduces sampling variability, though mitigated by robustness runs.
- **Interpretation as association, not causation:** spatially varying coefficients describe correlational patterns, not causal mechanisms. Confounding (e.g., unmeasured enforcement or street design) remains possible.
- **AADT imputation:** ~43% AADT missingness, with imputation to tract median, may introduce bias if missing areas differ systematically (e.g., local roads vs. state highways).
- **Temporal dynamics not modeled:** we do not account for changes in infrastructure, enforcement, or driving patterns over the 2019–2023 period.

6.3.3 Evaluation caveats

- **Modest discrimination:** even under tract-heldout evaluation, AUC 0.55–0.65, indicating that available predictors explain limited variance in severity. This suggests: (i) unmeasured factors (road geometry, occupant characteristics, speed) are important, or (ii) severity is partially stochastic.
- **Generalization to other regions:** Los Angeles is polycentric, sprawling, and diverse; findings may not generalize directly to compact cities, rural regions, or areas with different enforcement/infrastructure norms.

6.4 Recommendations for future work

1. **Replicate on SWITRS/TIMS:** conduct identical analysis on California’s official crash database to validate findings and assess data-source sensitivity.
 2. **Incorporate road-network data:** match crashes to roadway segments and extract functional class, speed limit, and turn-conflict geometry for more direct exposure/design controls.
 3. **Temporal and seasonal analysis:** extend MGWR framework to include year and season effects, capturing changes in driver behavior and infrastructure over time.
 4. **Occupant characteristics:** integrate vehicle type, occupant age/belt use (where available) to separate driver/vehicle factors from environmental factors.
 5. **Mechanistic validation:** conduct qualitative research (interviews, site visits) in neighborhoods with strong socio-economic-severity associations to understand local mechanisms.
 6. **Policy pilot testing:** use MGWR coefficient maps to target specific neighborhoods for pilot interventions (e.g., speed-management in high-income areas with strong vehicle-unavailability associations, enforcement escalation in auto-oriented corridors), then evaluate impact.
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7 Conclusion

This study demonstrates that **integrating machine learning feature screening with spatially varying coefficient modeling (MGWR) provides both predictive and interpretive value** for understanding crash severity. By combining Random Forest for variable importance with MGWR for spatially heterogeneous associations, we uncover place-specific mechanisms linking socio-economic vulnerability, built environment, and crash outcomes in Los Angeles.

Our findings support a **transportation-safety-as-equity framework**: crash severity is not randomly distributed; it clusters in neighborhoods where socio-economic vulnerability is highest. However, the **strength and direction** of these associations vary geographically, suggesting that intervention strategies must be tailored to local contexts rather than applied uniformly citywide.

Methodologically, this work advances the field by: (i) implementing rigorous spatial evaluation (tract-heldout) to avoid optimistic bias, (ii) demonstrating practical MGWR workflows with computational caching for resource-limited settings, and (iii) providing fully reproducible, open-source code enabling replication and adaptation by other practitioners.

The path forward is clear: this methodology should be replicated on SWITRS (California’s official crash database) for policy validation. Combining official crash data with MGWR’s local heterogeneity assessment could inform evidence-based, equitable resource allocation in traffic safety—directing investments in speed management, pedestrian infrastructure, and enforcement toward neighborhoods where need is greatest and local evidence is strongest.

Traffic safety is not a problem with a single global solution. It is a problem with many local solutions, waiting to be discovered through place-based analysis and policy.